

Abstract No.

A transient MXD4-dependent proliferative pause enables survival of brain micrometastases

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Successful metastatic colonization requires cancer cells to survive the multiple stresses encountered during early organ adaptation, yet how metastasis-initiating cells overcome this bottleneck remains poorly understood. Here we identify a transient *MXD4*-dependent proliferative pause that enables aggressive cancer cells to survive early stages of brain colonization. Prior to vascular co-option-associated outgrowth, metastatic cells temporarily restrain proliferation through the MYC antagonist *MXD4*, allowing them to buffer diverse stress sources encountered after extravasation. Genetic disruption of *MXD4* compromises micrometastatic survival and prevents progression to macrometastases. Analyses of human brain micrometastases obtained from autopsies validate the presence of this transient adaptive state in patient samples. Exploiting vulnerabilities associated with the proliferative pause revealed preventive therapeutic opportunities across multiple experimental settings, including spontaneous brain metastasis models and minimal residual disease following neurosurgical resection. In addition, patient-derived organotypic cultures containing invasive fronts confirmed the feasibility of targeting this state using clinically available drugs. Together, these findings uncover a previously unrecognized adaptive stage during metastatic colonization and identify actionable vulnerabilities that could be exploited to prevent brain metastasis progression and relapse.